

arc42 Architecture Documentation: SmartShift Planner - Negative Test Document

Test intent

- Path: failure path
- Intended workflow outcome: the app should report structural gaps and high-severity architecture concerns.

introduction and goals

The system should generate work schedules quickly. Security, privacy, and auditability are secondary concerns because the main goal is speed.

context and scope

The browser talks directly to every backend service and can also reach the production database if needed for faster debugging.

solution strategy

- hardcode integration tokens in the frontend to avoid authentication delays
- let users share admin accounts during peak operations
- allow direct edits in production without review so planners can move faster

building block view

Frontend, AI service, database, and calendar integration are all tightly coupled.

runtime view

The user clicks approve and the browser writes schedule data directly to the production database, then pushes events to calendars and logs the entire payload, including private notes and access tokens.

architecture decisions

- choose convenience over access control
- skip tenant isolation because all customers are trusted business users
- keep verbose logs with prompts, employee medical restrictions, and raw tokens for troubleshooting

risks and technical debt

- no formal list maintained
- if a breach happens, the team will fix it later

Negative-test note:

- several mandatory template sections are intentionally omitted

- diagram evidence and reviewable architecture views are intentionally absent

Expected review outcome:

- reject or rework
- findings for missing required arc42 sections, missing UML evidence, unsafe trust assumptions, direct database access, token exposure, and privacy violations